



SunBOT: Solar Cells That Track The Sun Like A Sunflower

Present-day commercial silicon solar cells have about 22 per cent efficiency. One of the most efficient ways to increase photovoltaic performance is integrating heliotropism, that is, the ability to track the Sun across the sky, using solar cells for more efficient photon collection



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The ability to follow the source of sunlight is called heliotropism (helio relates to the Sun and tropic to loving), whereas phototropism is the ability to track light. Sunlight causes destruction of a plant hormone group called auxins that build up in the cells on the shady side. Auxins, like indole acetic acid (IAA), promote cell growth and division. Hence, there is much more growth of longer cells and a higher number of cells on the shady side.

Uneven growth pattern makes one side longer than the other and forces the stalk to turn. As shown in Fig. 1, in the first and second sections when sunlight is overhead, auxins (IAA) molecules are evenly distributed in the stalk of the sunflower. Once sunlight shines at an angle, auxins move to the far side where there is shade, and induce elongation of cells on that side, as shown in the third and fourth sections of Fig. 1. As a result of cell elongation,

stalks bend towards the light, as shown in Fig. 1 (fourth section). Sunflowers are one of several plant species that make optimum use of light to increase growth rate.

Mimicking this natural phenomenon, scientists have now built artificial stems with special materials to support solar cells that can track sunlight, like sunflowers. These stem-like artificial materials can bend towards light, are self-regulating and move into the optimal position needed to absorb Sun's rays. Then, they make small adjustments to stay there as the Sun shifts. This allows them to capture about ninety per cent of available light shining on solar cells at a 75° angle.

Without tracking the Sun, loss can be enormous when incident angle is large (for example, 75 per cent loss at 75° incidence). These phototropic systems are termed as SunBOT, or sunflower-like biometric omnidirectional trackers, as shown in Fig. 2. These can autonomously and instantane-

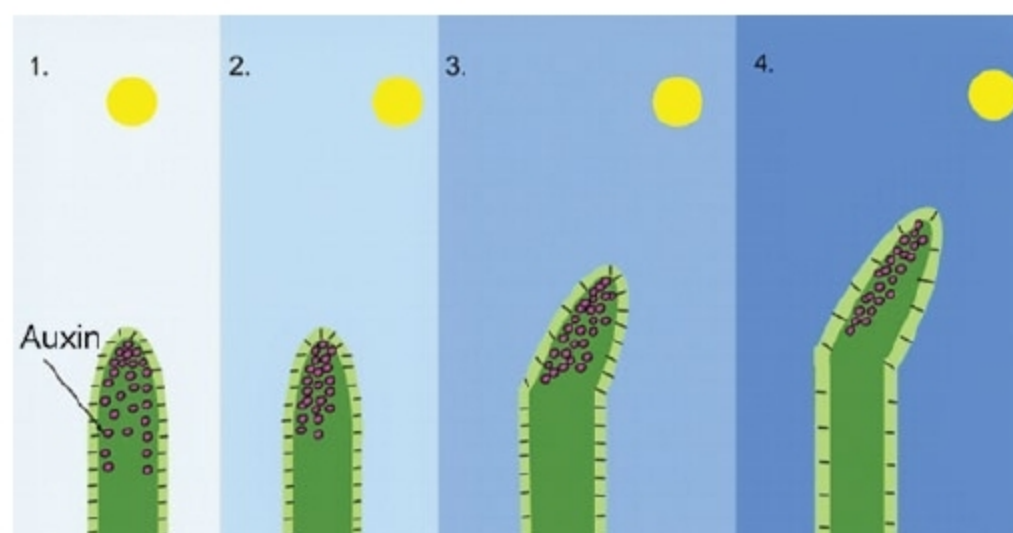


Fig. 1: Operating mechanism of heliotropism in sunflower



Fig. 2: Simile of sunflower (background) and SunBOT stem (foreground) tracking the sun